



Chapter 11: Probability

Three-week revision booklet · 6 worksheets · 108 questions

NSW syllabus outcome

MA4-PRO-C-01 — solves problems involving the probabilities of simple chance experiments · MAO-WM-01

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How to use this booklet

There are six worksheets, two for each week, and each one runs across **two pages**. Every section opens with a short **worked example** showing you exactly how that kind of question is done, and wherever a picture helps — a spinner, a die, a bag of counters or the probability scale — the diagram is drawn for you. Each sheet is split into **Easy**, **Medium** and **Challenging**, and questions that need working out are given ruled space for it. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Sample spaces
1	2	Basic probability
2	3	Fractions, decimals and percentages
2	4	Complementary events
3	5	Probability in context
3	6	Mixed revision

Progress tracker

Worksheet	Date completed	Score / 18	Checked by
1 Sample spaces			
2 Basic probability			
3 Fractions, decimals and percentages			
4 Complementary events			
5 Probability in context			
6 Mixed revision			

Worksheet 1: Sample spaces

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Week 1 · Worksheet 1 of 6 · 18 questions

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The sample space is the list of every possible outcome of an experiment.

EASY

9 QUESTIONS

WRITING A SAMPLE SPACE

- List **every** outcome once, in a sensible order.
- Rolling a die: the sample space is **1, 2, 3, 4, 5, 6** — that is 6 outcomes.
- Tossing a coin: **heads, tails** — 2 outcomes.

Write the sample space for each experiment.

1. Tossing a coin

2. Rolling a standard die

3. Spinning a spinner labelled 1, 2, 3

Answer: _____

Answer: _____

Answer: _____

4. Choosing a day of the weekend

5. Tossing a drawing pin (up or down)

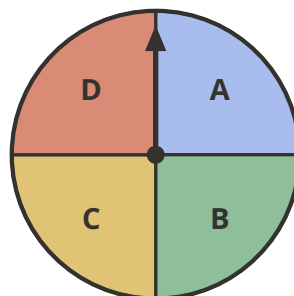
6. Choosing a season of the year

Answer: _____

Answer: _____

Answer: _____

The spinner has four equal sectors.



Four equal sectors

7. Write the sample space.

8. How many outcomes are there?

9. Are all four outcomes equally likely? Explain.

Answer: _____

Answer: _____

Answer: _____

Worksheet 1: Sample spaces

Week 1 · Worksheet 1 of 6 · continued

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MEDIUM

6 QUESTIONS

COUNTING OUTCOMES

- You do not always have to list them — sometimes you just count them.
- A full deck has 52 cards, so choosing one card has **52** outcomes.
- Outcomes are **equally likely** only if each has the same chance.

One counter is taken from the bag without looking.



3 red, 2 blue, 1 green

10. Write the sample space for the colour.

Answer:

11. How many counters are in the bag?

Answer:

12. Which colour is the most likely? Explain.

Answer:

How many outcomes are in each sample space?

13. Choosing a card from a full deck

Answer:

14. Choosing a letter of the alphabet

Answer:

15. Tossing two coins

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- For two-step experiments, set the outcomes out in a list or a table so none are missed.

Read each one, draw a diagram if it helps, then answer.

16. List the sample space for tossing a coin twice. Set your outcomes out in a list or a table so none are missed.

Draw your diagram here

Answer:

17. Design a spinner with four sectors where all outcomes are equally likely. Draw it and write its sample space.

Draw your diagram here

Answer:

18. Give an example of an event with probability 0 and one with probability 1. Explain each of your answers.

Draw your diagram here

Answer:

Worksheet 2: Basic probability

NAME _____

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$P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of outcomes}$. Give answers as fractions in simplest form.

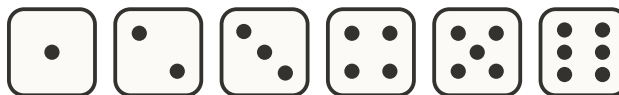
EASY

9 QUESTIONS

WORKING OUT A PROBABILITY

- Count the outcomes you **want**, then count the outcomes **altogether**.
- Rolling a die, $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$ — three even numbers out of six.
- Always simplify the fraction at the end.

A standard die is rolled. Find each probability.



A standard die

1. $P(6)$

2. $P(\text{an even number})$

3. $P(\text{a number greater than 2})$

Answer: _____

Answer: _____

Answer: _____

4. $P(1)$

5. $P(\text{an odd number})$

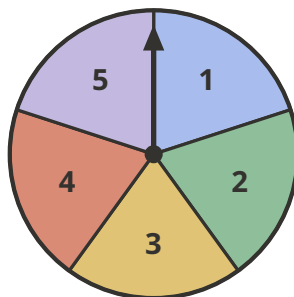
6. $P(\text{a multiple of 3})$

Answer: _____

Answer: _____

Answer: _____

The spinner is spun once. Find each probability.



Five equal sectors

7. $P(3)$

8. $P(\text{an even number})$

9. $P(\text{more than 3})$

Answer: _____

Answer: _____

Answer: _____

Worksheet 2: Basic probability

Week 1 · Worksheet 2 of 6 · continued

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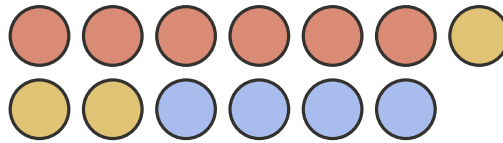
MEDIUM

6 QUESTIONS

"OR" AND "NOT"

- "Red **or** blue" means count the reds and the blues together.
- "**Not** red" means count everything that is not red.
- An outcome that cannot happen has probability **0**.

One counter is taken from the bag. Find each probability.



6 red, 3 yellow, 4 blue

10. $P(\text{red})$

11. $P(\text{not red})$

12. $P(\text{red or blue})$

Answer:

Answer:

Answer:

A card is drawn from a standard deck of 52. Find each probability.

13. $P(\text{a heart})$

14. $P(\text{an Ace})$

15. $P(\text{a red card})$

Answer:

Answer:

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Find the total number of outcomes first — that is the denominator of every answer.

Read each one, show your working, then answer.

16. A box has 15 dark, 13 milk and 12 white chocolates. Find $P(\text{white})$, giving your answer as a fraction in simplest form.

Draw your diagram here

Answer:

17. A bag has 6 red, 3 blue and 1 yellow counter. Find $P(\text{not red})$ and explain what that probability means.

Draw your diagram here

Answer:

18. Five names are in a hat: 3 girls and 2 boys. Find $P(\text{a girl's name})$ and $P(\text{one particular boy's name})$.

Draw your diagram here

Answer:

Worksheet 3: Fractions, decimals and percentages

Week 2 · Worksheet 3 of 6 · 18 questions

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The same probability can be written as a fraction, a decimal or a percentage.

EASY

9 QUESTIONS

CHANGING BETWEEN FORMS

- Fraction to decimal: divide the top by the bottom. $\frac{1}{4} = 1 \div 4 = 0.25$.
- Decimal to percentage: multiply by 100. $0.25 = 25\%$.
- Percentage to fraction: put it over 100 and simplify. $45\% = \frac{45}{100} = \frac{9}{20}$.

Write each probability as a decimal, then as a percentage.

1. $\frac{1}{2}$

2. $\frac{1}{4}$

3. $\frac{3}{4}$

Answer: _____

Answer: _____

Answer: _____

4. $\frac{1}{5}$

5. $\frac{2}{5}$

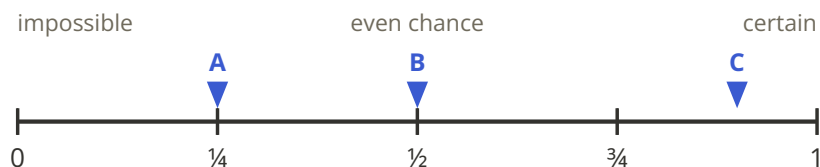
6. $\frac{7}{10}$

Answer: _____

Answer: _____

Answer: _____

Three events A, B and C are marked on the probability scale.



7. Write the probability of A as a fraction and a percentage.

8. Write the probability of B as a decimal.

9. Which event is the most likely? Describe it in words.

Answer: _____

Answer: _____

Answer: _____

Worksheet 3: Fractions, decimals and percentages

Week 2 · Worksheet 3 of 6 · continued

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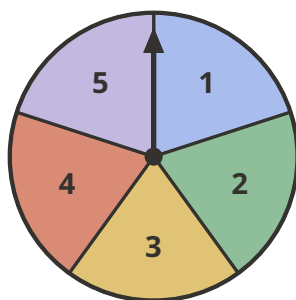
MEDIUM

6 QUESTIONS

COMPARING PROBABILITIES

- To compare, change them all into the **same form** — decimals are easiest.
- On the probability scale, further right means more likely.

The spinner is spun once. Give each answer three ways: as a fraction, a decimal and a percentage.



Five equal sectors

10. $P(4)$

Answer:

11. $P(\text{an odd number})$

Answer:

12. $P(\text{more than 2})$

Answer:

Convert as directed.

13. $P = 0.6$ as a percentage

Answer:

14. $P = 45\%$ as a decimal

Answer:

15. $P = \frac{3}{8}$ as a percentage

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- For "so many out of so many", write the fraction first, then convert it.

Read each one, show your working, then answer.

16. A survey of 500 people found 220 prefer coffee. Find $P(\text{coffee})$ as a percentage, correct to 1 decimal place.

Draw your diagram here

Answer:

17. Order these from least to most likely: 0.4, 35%, $\frac{1}{3}$. Show how you compared them.

Draw your diagram here

Answer:

18. A spinner has $P(\text{red}) = 25\%$. Out of 60 spins, how many reds would you expect? Explain your reasoning.

Draw your diagram here

Answer:

Worksheet 4: Complementary events

Week 2 · Worksheet 4 of 6 · 18 questions

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The complement of an event is everything else that could happen. $P(\text{event}) + P(\text{complement}) = 1$.

EASY

9 QUESTIONS

FINDING THE COMPLEMENT

- Take the probability away from 1 (or from 100%).
- $1 - \frac{1}{4} = \frac{3}{4} \cdot 100\% - 45\% = 55\% \cdot 1 - 0.3 = 0.7$
- The two probabilities always add to 1.

Find the complement of each probability.

1. $\frac{1}{4}$

2. $\frac{3}{5}$

3. $\frac{7}{10}$

Answer: _____

Answer: _____

Answer: _____

4. 0.3

5. 45%

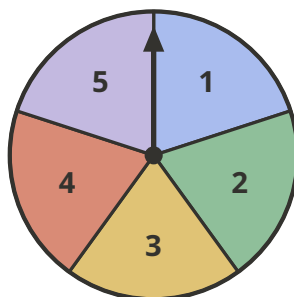
6. 0.85

Answer: _____

Answer: _____

Answer: _____

The spinner is spun once.



Five equal sectors

7. $P(\text{not } 3)$

8. $P(\text{not an even number})$

9. Write the complement of "spinning a 5" in words.

Answer: _____

Answer: _____

Answer: _____

Worksheet 4: Complementary events

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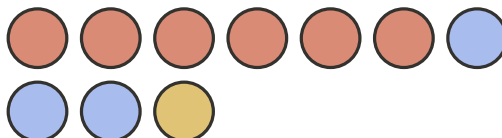
MEDIUM

6 QUESTIONS

COMPLEMENTS IN WORDS

- The complement of "rolling a 6" is "**not** rolling a 6".
- Say it in the clearest way you can: the complement of "it rains" is "it does not rain".

One counter is taken from the bag.



6 red, 3 blue, 1 yellow

10. $P(\text{not red})$

11. $P(\text{not blue})$

12. $P(\text{red}) + P(\text{not red})$. What do you notice?

Answer: _____

Answer: _____

Answer: _____

Write the complement of each event in words.

13. Choosing an Ace from a deck

14. Rolling a multiple of 3 on a die

15. A coin lands heads

Answer: _____

Answer: _____

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- It is often quicker to find $P(\text{event})$ first, then subtract from 1.

Read each one, show your working, then answer.

16. A golfer has a 74% chance of sinking a putt. Find $P(\text{missing})$, and give your answer as a decimal as well.

17. $P(\text{rain}) = 0.35$. Find $P(\text{no rain})$ as a percentage, and describe how likely no rain is in words.

18. Explain why $P(\text{event}) + P(\text{complement})$ must always equal 1. Use an example to show it.

Draw your diagram here

Draw your diagram here

Draw your diagram here

Answer: _____

Answer: _____

Answer: _____

Worksheet 5: Probability in context

Week 3 · Worksheet 5 of 6 · 18 questions

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Read carefully to find the total number of outcomes before you write the probability.

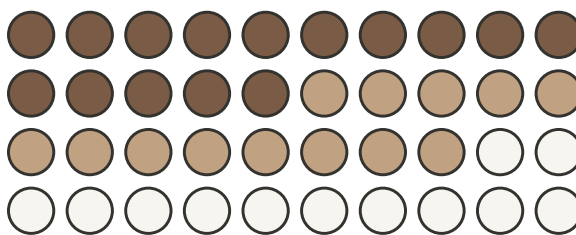
EASY

9 QUESTIONS

GETTING THE TOTAL RIGHT

- Add every group together to get the total number of outcomes.
- 15 dark + 13 milk + 12 white = **40** chocolates, so $P(\text{milk}) = \frac{13}{40}$.
- If something is not in the list at all, its probability is **0**.

A chocolate is taken from the box at random. Find each probability.



15 dark, 13 milk, 12 white chocolates

1. $P(\text{dark})$

2. $P(\text{milk})$

3. $P(\text{white})$

Answer: _____

Answer: _____

Answer: _____

4. $P(\text{not white})$

5. $P(\text{dark or milk})$

6. $P(\text{a caramel})$

Answer: _____

Answer: _____

Answer: _____

A truck carries 240 sanitiser, 305 soap, 335 paper and 120 pen boxes.

7. How many boxes altogether?

8. $P(\text{paper})$

9. $P(\text{not soap})$

Answer: _____

Answer: _____

Answer: _____

Worksheet 5: Probability in context

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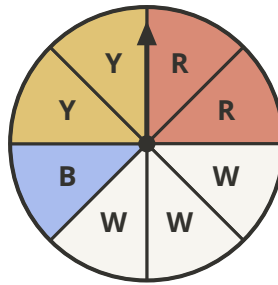
MEDIUM

6 QUESTIONS

EXPECTED AND OBSERVED

- **Theoretical** probability is worked out; **observed** comes from an experiment.
- Expected frequency = probability $\times$ number of trials.
- $P(1 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$, so in 60 rolls you would expect about **20**.

The 8-sided die is rolled once. Give each answer as a decimal.



An 8-sided die: 2 red, 3 white, 1 blue, 2 yellow

10. $P(\text{blue})$

Answer:

11. $P(\text{white or yellow})$

Answer:

12. $P(\text{not red})$

Answer:

Answer each question.

13. A die is rolled 80 times and 1 or 6 comes up 25 times. Find the observed probability as a percentage.

Answer:

14. Find the theoretical probability of rolling 1 or 6, as a percentage (1 dp).

Answer:

15. How many times would you expect 1 or 6 in 80 rolls?

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Write the total down first, then work out each probability from it.

Read each one, show your working, then answer.

16. A spinner has $P(\text{red}) = 30\%$. In 200 spins, how many reds would you expect? Explain your reasoning.

Draw your diagram here

Answer:

17. Explain the difference between theoretical and observed probability, using the die rolled 80 times as your example.

Draw your diagram here

Answer:

18. Design a spinner with four sectors where $P(A) = 50\%$ and $P(B) = P(C) = P(D)$. Draw it and label every sector.

Draw your diagram here

Answer:

Worksheet 6: Mixed revision

NAME _____

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This sheet mixes every skill from the Probability chapter. Work through it as though it were the topic test.

EASY

9 QUESTIONS

QUICK REMINDERS

- $P(\text{event}) = \text{favourable} \div \text{total}$. $P(\text{not } A) = 1 - P(A)$.
- $\frac{1}{4} = 0.25 = 25\%$. Probabilities always sit between 0 and 1.
- The sample space is the list of every possible outcome.

A standard die is rolled. Find each probability.



A standard die

1. $P(6)$

Answer: _____

2. $P(\text{an even number})$

Answer: _____

3. $P(\text{a multiple of 3})$

Answer: _____

The spinner is spun once.



Five equal sectors

4. Write the sample space.

Answer: _____

5. $P(3)$

Answer: _____

6. $P(\text{not an even number})$

Answer: _____

One counter is taken from the bag.



6 red, 3 blue, 1 yellow

7. $P(\text{red})$

Answer: _____

8. $P(\text{not red})$

Answer: _____

9. $P(\text{blue or yellow})$

Answer: _____

Worksheet 6: Mixed revision

Week 3 · Worksheet 6 of 6 · continued

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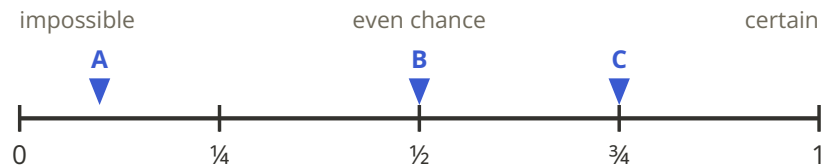
MEDIUM

6 QUESTIONS

CHOOSING WHAT TO DO

- Asked for “not”? Use the complement. Asked for “or”? Add the groups together.
- Expected frequency = probability $\times$ number of trials.

Three events are marked on the probability scale.



10. Write the probability of A as a percentage.

Answer: _____

11. Write the probability of C as a fraction.

Answer: _____

12. Describe event B in words.

Answer: _____

Answer each question.

13. A box has 15 dark, 13 milk and 12 white chocolates. Find $P(\text{white})$.

Answer: _____

14. A golfer sinks 74% of putts. Find $P(\text{missing})$.

Answer: _____

15. Find the expected number of 6s when a die is rolled 60 times.

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Find the total number of outcomes first, then work through the question step by step.

Read each one, show your working, then answer.

16. Which of these is certain: rolling a 7 on a die, drawing red from a bag of blue counters, or flipping heads or tails? Explain.

Draw your diagram here

Answer: _____

17. A survey of 500 people found 220 prefer coffee and 180 prefer tea. Find $P(\text{neither})$ as a percentage.

Draw your diagram here

Answer: _____

18. Explain why the probabilities of an event and its complement always add to 1. Use an example to show it.

Draw your diagram here

Answer: _____